

Name: Natalie 9.4.26 Hour: 1st

Tallest Tower Engineering Challenge

Materials: 8 straws + 5 paper clips

We will use the Engineering Design Process to solve this challenge.

1. ASK 🤔

Challenge: Build the tallest free-standing tower possible.

Constraints (rules):

- Use only 8 straws and 5 paper clips.
- The tower must stand on its own.

What other constraints might affect your design?

Standing on its own, gravity-balance, outside forces.

2. RESEARCH 📖

What makes towers stable?

1. A wide base

2. a strong foundation that safely spreads out weight

3. IMAGINE ?

Brainstorm at least 2 possible designs.

1. 	2. 
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Name: _____ Hour: _____

4. PLAN

Select the most promising design.

Why did you choose this design?

it has a wide base

How will you connect the straws with paper clips?

with the holes of the straws

5. CREATE

Build your prototype.

Sketch your final design here:



6. TEST

Trial Data Table

Trial	Tower Height (cm)	Stood for 10 seconds? (Y/N)	Notes
1	88cm	yes	it was a wide base
2	64	no	it wasn't a wide base
3			

Name: _____ Hour: _____

7. IMPROVE

What changes would make your tower taller or more stable?

Wider base.

Reflection (CER)

Claim: Was your design successful? yes (yes or no)

Evidence: Use your data table (height of tower?)

88 cm

Reasoning: Why did your design work (or not work)?

It had a wide base

Name: Leah Hour: 1st

Tallest Tower Engineering Challenge

Materials: 8 straws + 5 paper clips + tape measure

We will use the Engineering Design Process to solve this challenge.

1. ASK

Challenge: Build the tallest free-standing tower possible.

Constraints (rules):

- Use only 8 straws and 5 paper clips.
- The tower must stand on its own.

What other constraints might affect your design?

Standing on its own, fairly, balance, at least 5cm

2. RESEARCH Tip

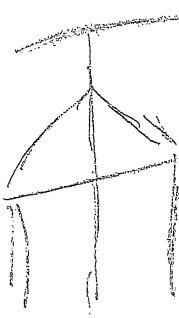
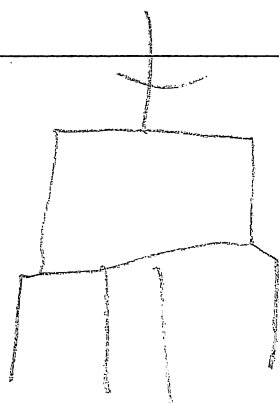
What makes towers stable?

1. Wide base
2. low center of gravity

3. IMAGINE ?

Brainstorm at least 2 possible designs.

*Sketch
of wide
or base*

1. 	2. 
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Name: _____ Hour: _____

4. PLAN

Select the most promising design.

Why did you choose this design?

① cuz we can get more height

How will you connect the straws with paper clips?

we will attach them

5. CREATE

building time

Build your prototype.

Sketch your final design here:

6. TEST

Trial Data Table

Trial	Tower Height (cm)	Stood for 10 seconds? (Y/N)	Notes
1	28 cm	NO	It wasnt even so it didnt balance
2	30 cm	YES	
3			

Name: _____ Hour: _____

7. IMPROVE

What changes would make your tower taller or more stable?

Maybe we could make a better base

Reflection (CER)

Claim: Was your design successful? yes (yes or no)

Evidence: Use your data table (height of tower?)

The second one we did wasn't tall but
sturdy

Reasoning: Why did your design work (or not work)?

too wobbly

Name: Charlie Soule 9.14.26 951
Hour: _____

Tallest Tower Engineering Challenge

Materials: 8 straws + 5 paper clips

We will use the Engineering Design Process to solve this challenge.

1. ASK

Challenge: Build the tallest free-standing tower possible.

Constraints (rules):

- Use only 8 straws and 5 paper clips.
- The tower must stand on its own.

What other constraints might affect your design?

standing on its own, gravity, seismic, outside forces.

2. RESEARCH

What makes towers stable?

1. wide base
2. a heavy base

3. IMAGINE

Brainstorm at least 2 possible designs.

1. 	2. 
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Name: Charlie Hour: _____

4. PLAN

Select the most promising design.

Why did you choose this design?

because it had a wide, sturdy base and

looked quite promising.

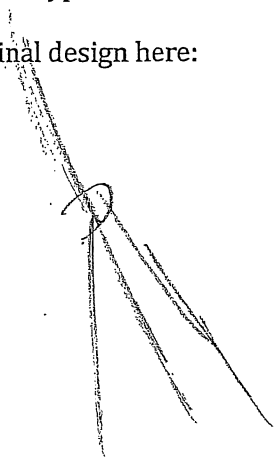
How will you connect the straws with paper clips?

stick them in them to make joints.

5. CREATE

Build your prototype.

Sketch your final design here:



6. TEST

Trial Data Table

Trial	Tower Height (cm)	Stood for 10 seconds? (Y/N)	Notes
1	77cm	Yes	move legs a little closer unstable
2	85cm	Yes	
3	83 $\frac{1}{2}$ cm	NO	

86 $\frac{1}{2}$ cm

Yes

Name: _____ Hour: _____

7. IMPROVE

What changes would make your tower taller or more stable?

Adding another paper clip

Reflection (CER)

Claim: Was your design successful? Yes (yes or no)

Evidence: Use your data table (height of tower?)

84 centimeters

Reasoning: Why did your design work (or not work)?

Our design works because we used the paper

clips to connect the straws like joints.

Name: Dylan 9.14.21 Hour: 2~~nd~~ hr

Tallest Tower Engineering Challenge

Materials: 8 straws + 5 paper clips + tape measures

We will use the Engineering Design Process to solve this challenge.

1. ASK

Challenge: Build the tallest free-standing tower possible.

Constraints (rules):

- Use only 8 straws and 5 paper clips.
- The tower must stand on its own.

What other constraints might affect your design?

Connections btwn straws, good base, balance

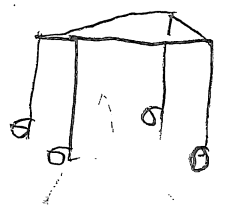
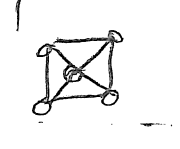
2. RESEARCH

What makes towers stable?

1. have more height at the bottom
2. use strong shapes like triangles

3. IMAGINE

Brainstorm at least 2 possible designs.

1. 	2. 
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Name: _____ Hour: _____

4. PLAN

Select the most promising design.

Why did you choose this design?

to make it stable and tall

How will you connect the straws with paper clips?

Poking wide through them

5. CREATE

Build your prototype.

Sketch your final design here:



6. TEST

Trial Data Table

Trial	Tower Height (cm)	Stood for 10 seconds? (Y/N)	Notes
1 Jimmy	51	yes	Make it + Stable
2 Jimmy 2.0	89	yes	↓
3			

Name: Dylan Hour: _____

7. IMPROVE

What changes would make your tower taller or more stable?

Plating a straw in to another straw (makes taller)

Reflection (CER)

Claim: Was your design successful? Yes (yes or no)

Evidence: Use your data table (height of tower?)

the height of the base / Jimmy

Reasoning: Why did your design work (or not work)?

because we made it stable at the bottom

